Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Order of Operations in Algebra

When an algebraic expression contains several operations, you must do them in a fixed order so that everyone gets the same answer. Work out any **brackets** first, then do **multiplication and division** from left to right, and finally **addition and subtraction** from left to right.

In algebra, remember that a term like $3x$ means $3 \times x$. So $3x \times 2$ really means $3 \times x \times 2$, and $17y - 5 \times 2y$ means you must do the multiplication $5 \times 2y$ before you subtract. Likewise, $7x \div x$ means $7 \times x \div x$, which simplifies because the x's cancel.

Example 1 — finding values in order

(a) $18 - 6 \times 2$

Multiply first: $6 \times 2 = 12$, then subtract.

$18 - 12 = 6$

(b) $6 - 18 \div 3$

Divide first: $18 \div 3 = 6$, then subtract.

$6 - 6 = 0$

(c) $6 \times 2 - 2 - 40 \div 4$

Do the multiplication and division first: $6 \times 2 = 12$ and $40 \div 4 = 10$, then subtract left to right.

$12 - 2 - 10 = 10 - 10 = 0$

Example 2 — simplifying expressions

(a) $3x \times 2 + 5x$

Multiply first: $3x \times 2 = 6x$. Then add the like terms.

$6x + 5x = 11x$

(b) $17y - 5 \times 2y$

Multiply first: $5 \times 2y = 10y$. Then subtract the like terms.

$17y - 10y = 7y$

Example 3 — division and brackets

(a) $7x \div x + 5$

Divide first: $7x \div x = 7$ (the x's cancel). Then add.

$7 + 5 = 12$

(b) $4x - (6 \div 3)$

Work inside the bracket first: $6 \div 3 = 2$. Then subtract.

$4x - 2$

(c) $(24x \div 6) + x + (1 \times 5x)$

Work the brackets first: $24x \div 6 = 4x$ and $1 \times 5x = 5x$. Then add the like terms.

$4x + x + 5x = 5x + 5x = 10x$

Practice

1. Simplify $4n + 3n \times 10$
2. Simplify $32 + 8x \div 2$

3. Simplify $3 \times 5x + 4x \div 2$
4. Simplify $36y \div 6 + y + 2 \times 4y$

Section 2: Removing Brackets and Simplifying Expressions

When an algebraic expression has brackets, you often want to remove them so you can collect like terms. Two rules tell you exactly what happens when the bracket comes off.

Rule 1 — a + sign in front of a bracket. A plus sign before a bracket does not change any sign inside the bracket. Every term inside keeps its own sign as the bracket is removed.

Rule 2 — a − sign in front of a bracket. A minus sign before a bracket changes every sign inside it: each + becomes −, and each − becomes +.

Before you start, always check whether the terms inside the bracket can be simplified first — if they can, do that first, then remove the bracket.

Example 1 — removing a bracket after a +

Simplify 6x + (3 - x).

The + before the bracket keeps the signs unchanged:
6x + (3 - x) = 6x + 3 - x

Now collect the like terms:
6x - x + 3 = 5x + 3

So 6x + (3 - x) = 5x + 3.

Example 2 — removing a bracket after a −

Simplify 10d - (9c - 4d).

The − before the bracket flips every sign inside: +9c becomes -9c and -4d becomes +4d.
10d - (9c - 4d) = 10d - 9c + 4d

Collect the like terms:
10d + 4d - 9c = 14d - 9c

So 10d - (9c - 4d) = 14d - 9c.

Example 3 — two brackets

Simplify (6x - y) - (4x - 5y).

Remove the first bracket: the + in front of it keeps signs unchanged, so it stays 6x - y.

Remove the second bracket: the − flips each sign, so 4x becomes -4x and -5y becomes +5y.
6x - y - 4x + 5y

Collect the like terms:
6x - 4x + 5y - y = 2x + 4y = 2(x + 2y)

So (6x - y) - (4x - 5y) = 2x + 4y, or 2(x + 2y).

Practice

1. Simplify $(12a - 7) + (4 - 5a)$.
2. Simplify $(8x - 2y) - (9x - 4y)$.
3. Simplify $12 - (7 - 2f) - (3 - f)$.
4. Simplify $(8a - 5) + (14 - 3a)$.

Section 3: Turning Word Problems into Algebraic Expressions

A word problem describes a situation in words, and your job is to turn those words into an algebraic expression. To do this, you decide what each unknown letter stands for, then use the words to build an expression step by step.

Two phrases come up again and again:

"...less than a..." means you subtract. If a pen costs P naira and a pencil costs 15 naira less, the pencil costs ₦(P − 15).
"...more than a..." means you add. If a pencil costs x naira and a rubber costs 40 naira more, the rubber costs ₦(x + 40).

Once you have each separate cost as an expression, you add them together and then simplify by collecting like terms. When an item "joins" a group, add it; when an item "leaves", subtract it.

Example 1

A pen costs ₦P and a pencil costs 15 naira less than a pen.

(a) Find the cost of a pencil.
(b) Find the cost of a pen and a pencil.

Solution

Cost of a pen = ₦P

(a) A pencil is 15 naira less than a pen, so subtract 15:
cost of a pencil = ₦(P − 15)

(b) To find both together, add the two costs:
cost of a pen and a pencil = ₦P + ₦(P − 15)
= ₦P + ₦P − ₦15
= ₦2P − ₦15
= ₦(2P − 15)

Example 2

A pencil costs ₦x and a rubber costs 40 naira more than a pencil.

(a) Find the cost of a rubber.
(b) Find the cost of a pencil and a rubber.

Solution

(a) Cost of a pencil = ₦x. A rubber is 40 naira more, so add 40:
cost of a rubber = ₦(x + 40)

(b) Add the two costs:
cost of a pencil and a rubber = ₦x + ₦(x + 40)
= ₦(x + x + 40)
= ₦(2x + 40)

Example 3

In a class there are x boys and y girls. This semester, 5 boys join and 3 girls leave the class. How many students are now in this class?

Solution

Total number = number of boys + number of girls.

5 boys join, so the number of boys becomes x + 5.
3 girls leave, so the number of girls becomes y − 3.

Total number = (x + 5) + (y − 3)
= x + y + 5 − 3
= x + y + 2

So there are now (x + y + 2) students in the class.

Practice

1. A pen costs h naira and a crayon costs ₦20 less than a pen.
   (a) Find the cost of a crayon.
   (b) Find the cost of a crayon and a pen.
2. Envelopes cost d naira per pack. A notepad costs 20 naira more than a pack of envelopes. Find the total cost of a pack of envelopes and a notepad.
3. A pen and a pencil together cost ₦x. A pen costs ₦y. Find the cost of 3 pencils.

Check your work

Attempt every question before you read this page.

Order of Operations in Algebra

1. $34n$

   Multiply first: $3n \times 10 = 30n$. Then add the like terms: $4n + 30n = 34n$.
2. $32 + 4x$

   Divide first: $8x \div 2 = 4x$. Then add: $32 + 4x$. The terms are not alike, so they cannot be combined further.
3. $17x$

   Do the multiplication and division first: $3 \times 5x = 15x$ and $4x \div 2 = 2x$. Then add the like terms: $15x + 2x = 17x$.
4. $15y$

   Do the division and multiplication first: $36y \div 6 = 6y$ and $2 \times 4y = 8y$. Then add the like terms: $6y + y + 8y = 15y$.

Removing Brackets and Simplifying Expressions

1. $7a - 3$

   Remove the bracket: the + before it keeps every sign unchanged, so $12a - 7 + 4 - 5a$. Collect like terms: $12a - 5a = 7a$ and $-7 + 4 = -3$. So the expression is $7a - 3$.
2. $-x + 2y$

   Remove the first bracket (a + before it): $8x - 2y$. Remove the second bracket (a − before it): each sign flips, so $-(9x - 4y)$ becomes $-9x + 4y$. Together: $8x - 2y - 9x + 4y$. Collect like terms: $8x - 9x = -x$ and $-2y + 4y = 2y$. So the expression is $-x + 2y$.
3. $2 + 3f$

   Remove the first bracket (a − before it): $12 - 7 + 2f$. Remove the second bracket (also a − before it): $-(3 - f)$ becomes $-3 + f$. Together: $12 - 7 + 2f - 3 + f$. Collect like terms: the numbers give $12 - 7 - 3 = 2$, and the f terms give $2f + f = 3f$. So the expression is $2 + 3f$.
4. $5a + 9$

   Remove the bracket (a + before it keeps every sign unchanged): $8a - 5 + 14 - 3a$. Collect like terms: $8a - 3a = 5a$ and $-5 + 14 = 9$. So the expression is $5a + 9$.

Turning Word Problems into Algebraic Expressions

1. (a) ₦(h − 20) (b) ₦(2h − 20)

   A crayon is 20 naira less than a pen, so subtract 20 from the pen's cost: crayon = ₦(h − 20). Together, pen + crayon = ₦h + ₦(h − 20) = ₦h + ₦h − ₦20 = ₦2h − ₦20 = ₦(2h − 20).
2. ₦(2d + 20)

   A notepad is 20 naira more than a pack of envelopes, so it costs ₦(d + 20). Total = pack of envelopes + notepad = ₦d + ₦(d + 20) = ₦(d + d + 20) = ₦(2d + 20).
3. ₦3(x − y)

A pen and a pencil cost ₦x, and the pen costs ₦y, so one pencil costs ₦x − ₦y = ₦(x − y). Three pencils cost 3 × ₦(x − y) = ₦3(x − y).

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-152900ac	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____